Second-order adjoint methods for frictional parameters in earthquake modeling

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1 Introduction

We study adjoint-based inversion for rate-state frictional parameters in earthquake modeling, as done in [1], and extend the analysis to the second-order adjoint, as done in [2], [3]. To simplify, we start by assuming no changes in fault normal stress. We also consider the initial state Ψ_0 as known (should be easy to generalize later).

2 Derivations

2.1 Forward and adjoint problems

From [1], we have the following systems:
Forward problem

$$\begin{aligned}
\rho \ddot{u}_J &= \partial_I C_{IJKL} \partial_K u_L + Q_J, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\overline{u} &= \overline{u}_0, \quad \dot{\overline{u}} = \overline{v}_0, && \overline{x} \in \Omega, && t = 0, \\
L\overline{u} &= \overline{g}, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
V_n &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\overline{T}^+ &= -\overline{T}^-, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\tau_J &= F_J(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\dot{\Psi} &= G(\overline{V}, \Psi, p), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Psi &= \Psi_0, && \overline{x} \in \Gamma, && t = 0.
\end{aligned} \tag{1}$$

Adjoint problem

$$\begin{aligned}
\rho \ddot{u}^\dagger_J &= \partial_I C_{IJKL} \partial_K u^\dagger_L + Q^\dagger_J, && \overline{x} \in \Omega, && t \in \mathcal{T}, \\
\overline{u}^\dagger &= \overline{0}, \quad \dot{\overline{u}}^\dagger = \overline{0}, && \overline{x} \in \Omega, && t = T, \\
L^\dagger \overline{u}^\dagger &= 0, && \overline{x} \in \partial\Omega, && t \in \mathcal{T}, \\
V^\dagger_n &= 0, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\overline{T}^{\dagger +} &= -\overline{T}^{\dagger -}, && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\tau^\dagger_J &= F^\dagger_J(\overline{V}^\dagger, \Psi^\dagger), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
-\dot{\Psi}^\dagger &= G^\dagger(\overline{V}^\dagger, \Psi^\dagger), && \overline{x} \in \Gamma, && t \in \mathcal{T}, \\
\Psi^\dagger &= 0, && \overline{x} \in \Gamma, && t = T.
\end{aligned} \tag{2}$$

The adjoint source $Q_J^\dagger$ is the residual (the difference between model output at receivers and data) injected at the receiver locations. The adjoint friction law and state evolution are given by

$$\begin{aligned} F_J^\dagger(\overline{V}^\dagger, \Psi^\dagger) &= \frac{\partial F_I}{\partial V_J} V_I^\dagger + \frac{\partial G}{\partial V_J}\Psi^\dagger, \\ G^\dagger(\overline{V}^\dagger, \Psi^\dagger) &= \frac{\partial F_J}{\partial \Psi} V_J^\dagger + \frac{\partial G}{\partial \Psi}\Psi^\dagger. \end{aligned} \tag{3}$$

The adjoint problem is linear, with coefficients that depend on the forward solution. One way to view the derivation is that we linearize the forward problem around the forward solution and then take the adjoint.

2.2 Second-order Lagrangian

The cost functional for the second-order problem is constructed by convolving the gradient (5.9 from [1]) with a fixed model perturbation in the parameter p_1. We here denote the perturbation Δp_1.

$$\mathcal{F}_2 = -\left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1\right)_{\Gamma\times\mathcal{T}}. \tag{4}$$

We construct the Lagrangian by adding the forward and adjoint problems [3]. The forward problem (FP) is multiplied by Lagrange multipliers $\Delta u_j^\dagger, \Delta\Psi^\dagger$ (the second-order adjoint solution) and the adjoint problem (AP) by $\Delta u_j, \Delta\Psi$ (the second-order forward solution).

$$\begin{aligned} \mathcal{L}_2 = &-\left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\ &+ \left(\Delta\dot{u}_j^\dagger, \rho\ddot{u}_j - \partial_i C_{ijkl}\partial_k u_l - Q_j\right)_{\Omega\times\mathcal{T}} + \left(\Delta\Psi^\dagger, \dot{\Psi} - G\right)_{\Gamma\times\mathcal{T}} \\ &+ \left(\Delta\dot{u}_j, \rho\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k u_l^\dagger - Q_j^\dagger\right)_{\Omega\times\mathcal{T}} + \left(\Delta\Psi, -\dot{\Psi}^\dagger - G^\dagger\right)_{\Gamma\times\mathcal{T}} \end{aligned} \tag{5}$$

Varying the second-order variables straightforwardly leads to the forward and adjoint problems. Before varying the forward and adjoint variables, we use integration by parts to rewrite the Lagrangian. From the derivations in [1], we know that

$$\begin{aligned} &\left(\dot{u}_j^\dagger, \rho\ddot{u}_j - \partial_i C_{ijkl}\partial_k u_l - Q_j\right)_{\Omega\times\mathcal{T}} + \left(\Psi^\dagger, \dot{\Psi} - G\right)_{\Gamma\times\mathcal{T}} \\ = &-\left(\rho\dot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k u_l, \dot{u}_j\right)_{\Omega\times\mathcal{T}} - \left(\dot{u}_j^\dagger, Q_j\right)_{\Omega\times\mathcal{T}} \\ &- \left(V_j^\dagger, F_j\right)_{\Gamma\times\mathcal{T}} + \left(F_j^\dagger, V_j\right)_{\Gamma\times\mathcal{T}} + \underbrace{\left(V_n^\dagger, \sigma_n\right)_{\Gamma\times\mathcal{T}}}_{\text{Ignore, assume no change in } \sigma_n} \\ &- \left(\Psi^\dagger, \Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Psi^\dagger, G\right)_{\Gamma\times\mathcal{T}} + \underbrace{\left(\Psi_0^\dagger, \Psi_0\right)_\Gamma}_{\text{Ignore for now}} \end{aligned} \tag{6}$$

The derivations assume initial and terminal conditions, correct adjoint boundary conditions on exterior boundaries, and continuity of forward and adjoint traction across the fault. Using (6) on the terms in (5), and thereby assuming homogeneous initial/terminal

conditions, allows us to rewrite $\mathcal{L}_2$ as

$$\begin{aligned}
\mathcal{L}_2 = &- \left(\Psi^\dagger \frac{\partial G}{\partial p_1} + V_j^\dagger \frac{\partial F_j}{\partial p_1}, \Delta p_1 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\rho \Delta \ddot{u}_j^\dagger - \partial_i C_{ijkl} \partial_k \Delta u_l^\dagger, \dot{u}_j \right)_{\Omega\times\mathcal{T}} - \left(\Delta \dot{u}_j^\dagger, Q_j \right)_{\Omega\times\mathcal{T}} \\
&- \left(\Delta V_j^\dagger, F_j \right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, V_j \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \dot{\Psi}^\dagger, \Psi \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \Psi^\dagger, G \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\rho \Delta \ddot{u}_j - \partial_i C_{ijkl} \partial_k \Delta u_l, \dot{u}_j^\dagger \right)_{\Omega\times\mathcal{T}} - \left(\Delta \dot{u}_j, Q_j^\dagger \right)_{\Omega\times\mathcal{T}} \\
&- \left(\Delta V_j, F_j^\dagger \right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j, V_j^\dagger \right)_{\Gamma\times\mathcal{T}} + \left(\Delta \dot{\Psi}, \Psi^\dagger \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \Psi, G^\dagger \right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{7}$$

As expected, this closely resembles equation (5.21) in [1]. Note that we disregard the Ψ_0-terms to make it more concise.

2.3 Variations

Varying the parameter p_2 leads to

$$\begin{aligned}
\partial_{p_2}\mathcal{L}_2 = &- \left(V_j^\dagger \frac{\partial^2 F_j}{\partial p_1 \partial p_2} + \Psi^\dagger \frac{\partial^2 G}{\partial p_1 \partial p_2}, \Delta p_1 \delta p_2 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\Delta V_j^\dagger, \frac{\partial F_j}{\partial p_2} \delta p_2 \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \Psi^\dagger, \frac{\partial G}{\partial p_2} \delta p_2 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\Delta V_j, \frac{\partial F_j^\dagger}{\partial p_2} \delta p_2 \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \Psi, \frac{\partial G^\dagger}{\partial p_2} \delta p_2 \right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{8}$$

The partial derivatives of the adjoint friction law and state evolution functions are

$$\begin{aligned}
\frac{\partial F_j^\dagger}{\partial p} &= \frac{\partial^2 F_i}{\partial V_j \partial p} V_i^\dagger + \frac{\partial^2 G}{\partial V_j \partial p} \Psi^\dagger, \\
\frac{\partial G^\dagger}{\partial p} &= \frac{\partial^2 F_j}{\partial \Psi \partial p} V_j^\dagger + \frac{\partial^2 G}{\partial \Psi \partial p} \Psi^\dagger,
\end{aligned} \tag{9}$$

which leads to

$$\begin{aligned}
\partial_{p_2}\mathcal{L}_2 = &- \left(V_j^\dagger \frac{\partial^2 F_j}{\partial p_1 \partial p_2} + \Psi^\dagger \frac{\partial^2 G}{\partial p_1 \partial p_2}, \Delta p_1 \delta p_2 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\Delta V_j^\dagger \frac{\partial F_j}{\partial p_2}, \delta p_2 \right)_{\Gamma\times\mathcal{T}} - \left(\Delta \Psi^\dagger \frac{\partial G}{\partial p_2}, \delta p_2 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(V_j^\dagger \left[\frac{\partial^2 F_j}{\partial V_i \partial p_2} \Delta V_i + \frac{\partial^2 F_j}{\partial \Psi \partial p_2} \Delta \Psi \right], \delta p_2 \right)_{\Gamma\times\mathcal{T}} \\
&- \left(\Psi^\dagger \left[\frac{\partial^2 G}{\partial V_i \partial p_2} \Delta V_i + \frac{\partial^2 G}{\partial \Psi \partial p_2} \Delta \Psi \right], \delta p_2 \right)_{\Gamma\times\mathcal{T}}.
\end{aligned} \tag{10}$$

Now, we seek the second-order forward problem and second-order adjoint problem. We start by varying the Lagrangian (7) with respect to the adjoint solution, which will lead to the second-order forward problem. Note that we are essentially taking the adjoint of the

adjoint problem here, since we are repeating the steps that led to the first-order adjoint equations, but with the adjoint solution playing the role of the forward solution and the second-order forward solution playing the role of the adjoint solution. The variation with respect to the adjoint solution reads

$$\begin{aligned}\delta\mathcal{L}_2 = &-\left(\delta\Psi^\dagger\frac{\partial G}{\partial p_1}+\delta V_j^\dagger\frac{\partial F_j}{\partial p_1},\Delta p_1\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\Delta V_j,\delta F_j^\dagger\right)_{\Gamma\times\mathcal{T}}+\left(\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &+\left(\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}-\left(\Delta\Psi,\delta G^\dagger\right)_{\Gamma\times\mathcal{T}}.\end{aligned}\tag{11}$$

The required variations of $F_j^\dagger$ and $G^\dagger$ are

$$\begin{aligned}\delta F_j^\dagger &= \frac{\partial F_i}{\partial V_j}\delta V_i^\dagger+\frac{\partial G}{\partial V_j}\delta\Psi^\dagger,\\ \delta G^\dagger &= \frac{\partial F_j}{\partial\Psi}\delta V_j^\dagger+\frac{\partial G}{\partial\Psi}\delta\Psi^\dagger.\end{aligned}\tag{12}$$

Substituting in (11) and rearranging terms yields

$$\begin{aligned}\delta\mathcal{L}_2 = &-\left(\delta\Psi^\dagger\frac{\partial G}{\partial p_1}+\delta V_j^\dagger\frac{\partial F_j}{\partial p_1},\Delta p_1\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\Delta V_j,\frac{\partial F_i}{\partial V_j}\delta V_i^\dagger+\frac{\partial G}{\partial V_j}\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}+\left(\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &+\left(\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}-\left(\Delta\Psi,\frac{\partial F_j}{\partial\Psi}\delta V_j^\dagger+\frac{\partial G}{\partial\Psi}\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}\\ = &-\left(\rho\Delta\ddot{u}_j-\partial_i C_{ijkl}\partial_k\Delta u_l,\delta\dot{u}_j^\dagger\right)_{\Omega\times\mathcal{T}}\\ &-\left(\frac{\partial F_j}{\partial p_1}\Delta p_1+\Delta V_i\frac{\partial F_j}{\partial V_i}+\Delta\Psi\frac{\partial F_j}{\partial\Psi}-\Delta F_j,\delta V_j^\dagger\right)_{\Gamma\times\mathcal{T}}\\ &-\left(\frac{\partial G}{\partial p_1}\Delta p_1+\Delta V_j\frac{\partial G}{\partial V_j}+\Delta\Psi\frac{\partial G}{\partial\Psi}-\Delta\dot{\Psi},\delta\Psi^\dagger\right)_{\Gamma\times\mathcal{T}}\end{aligned}\tag{13}$$

The variation is zero if the second-order solution solves the following problem.

Second-order forward problem

$$\begin{aligned}\rho\Delta\ddot{u}_J &= \partial_I C_{IJKL}\partial_K\Delta u_L, && \overline{x}\in\Omega, && t\in\mathcal{T},\\ \Delta\overline{u}=\overline{0},\quad \Delta\dot{\overline{u}}&=\overline{0}, && \overline{x}\in\Omega, && t=0,\\ L\Delta\overline{u} &= \overline{0}, && \overline{x}\in\partial\Omega, && t\in\mathcal{T},\\ \Delta V_n &= 0, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\ \Delta\overline{T}^+ &= -\Delta\overline{T}^-, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\ \Delta\tau_J &= \Delta F_J, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\ \Delta\dot{\Psi} &= \Delta G, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\ \Delta\Psi &= 0, && \overline{x}\in\Gamma, && t=0.\end{aligned}\tag{14}$$

where

$$\begin{aligned}
\Delta F_J &= \frac{\partial G^\dagger}{\partial V_J^\dagger}\Delta\Psi + \frac{\partial F_I^\dagger}{\partial V_J^\dagger}\Delta V_I + \frac{\partial F_J}{\partial p_1}\Delta p_1 \\
&= \frac{\partial F_J}{\partial \Psi}\Delta\Psi + \frac{\partial F_J}{\partial V_I}\Delta V_I + \frac{\partial F_J}{\partial p_1}\Delta p_1 \\
\Delta G &= \frac{\partial G^\dagger}{\partial \Psi^\dagger}\Delta\Psi + \frac{\partial F_J^\dagger}{\partial \Psi^\dagger}\Delta V_J + \frac{\partial G}{\partial p_1}\Delta p_1 \\
&= \frac{\partial G}{\partial \Psi}\Delta\Psi + \frac{\partial G}{\partial V_J}\Delta V_J + \frac{\partial G}{\partial p_1}\Delta p_1
\end{aligned} \tag{15}$$

The second-order forward problem is the linearized forward problem, but with different source terms. This is consistent with [3]. It is the adjoint of the adjoint problem, which in turn was the adjoint of the linearized forward problem, so after taking the adjoint twice we are back to the linearized forward problem. Unlike the adjoint problem, there is no wavefield source at receiver locations. Instead, the source terms appear in the friction law and state evolution. They depend on the forward solution and the model perturbation Δp_1. There is no dependence on the adjoint solution. The second-order forward solution appears to be the (first-order) change in the forward solution that the model perturbation Δp_1 would cause.

To derive the second-order adjoint problem, we vary the forward solution, yielding

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\Psi^\dagger\left(\frac{\partial^2 G}{\partial p_1 \partial V_j}\delta V_j + \frac{\partial^2 G}{\partial p_1 \partial \Psi}\delta\Psi\right) + V_j^\dagger\left(\frac{\partial^2 F_j}{\partial p_1 \partial V_j}\delta V_j + \frac{\partial^2 F_j}{\partial p_1 \partial \Psi}\delta\Psi\right), \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\
&- \left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k \Delta u_l^\dagger, \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}} \\
&- \left(\Delta V_j^\dagger, \delta F_j\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, \delta G\right)_{\Gamma\times\mathcal{T}} \\
&- \left(\Delta\dot{u}_j, \delta Q_j^\dagger\right)_{\Omega\times\mathcal{T}} \\
&- \left(\Delta V_j, \delta F_j^\dagger\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, \delta G^\dagger\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{16}$$

There are more terms than when varying the adjoint solution, because $F_j^\dagger$, $G^\dagger$ and $Q_j^\dagger$ all depend on the forward solution, whereas F_j, G and Q_j are independent of the adjoint solution. Recall that, for a velocity misfit, the adjoint source is

$$Q_j^\dagger(t) = \sum_{k=1}^{N_{rec}} -(\dot{u}_j(\overline{x}_r^{(k)}, t) - d_j(t))\hat{\delta}(\overline{x} - \overline{x}_r^{(k)}) \tag{17}$$

The required variations are (need better notation for different variations)

$$\begin{aligned}
\delta F_j &= \frac{\partial F_j}{\partial V_i}\delta V_i + \frac{\partial F_j}{\partial \Psi}\delta\Psi \\
\delta G &= \frac{\partial G}{\partial V_j}\delta V_j + \frac{\partial G}{\partial \Psi}\delta\Psi \\
\delta F_j^\dagger &= \frac{\partial F_j^\dagger}{\partial V_i}\delta V_i + \frac{\partial F_j^\dagger}{\partial \Psi}\delta\Psi \\
\delta G^\dagger &= \frac{\partial G^\dagger}{\partial V_i}\delta V_i + \frac{\partial G^\dagger}{\partial \Psi}\delta\Psi \\
\delta Q_j^\dagger &= \sum_{k=1}^{N_{rec}} -\delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)})
\end{aligned} \tag{18}$$

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\Psi^\dagger\left(\frac{\partial^2 G}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 G}{\partial p_1\partial\Psi}\delta\Psi\right) + V_j^\dagger\left(\frac{\partial^2 F_j}{\partial p_1\partial V_j}\delta V_j + \frac{\partial^2 F_j}{\partial p_1\partial\Psi}\delta\Psi\right), \Delta p_1\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k\Delta u_l^\dagger - \sum_{k=1}^{N_{rec}}\Delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)}), \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}} \\
&-\left(\Delta V_j^\dagger, \frac{\partial F_j}{\partial V_i}\delta V_i + \frac{\partial F_j}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} + \left(\Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi^\dagger, \frac{\partial G}{\partial V_j}\delta V_j + \frac{\partial G}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\Delta V_j, \frac{\partial F_j^\dagger}{\partial V_i}\delta V_i + \frac{\partial F_j^\dagger}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}} - \left(\Delta\Psi, \frac{\partial G^\dagger}{\partial V_i}\delta V_i + \frac{\partial G^\dagger}{\partial\Psi}\delta\Psi\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{19}$$

Gathering terms leads to

$$\begin{aligned}
\delta\mathcal{L}_2 = &-\left(\rho\Delta\ddot{u}_j^\dagger - \partial_i C_{ijkl}\partial_k\Delta u_l^\dagger - \sum_{k=1}^{N_{rec}}\Delta\dot{u}_j(\overline{x}_r^{(k)},t)\hat{\delta}(\overline{x}-\overline{x}_r^{(k)}), \delta\dot{u}_j\right)_{\Omega\times\mathcal{T}} \\
&-\left(\left(\Psi^\dagger\frac{\partial^2 G}{\partial p_1\partial\Psi} + V_j^\dagger\frac{\partial^2 F_j}{\partial p_1\partial\Psi}\right)\Delta p_1 + \frac{\partial F_j}{\partial\Psi}\Delta V_j^\dagger + \frac{\partial G}{\partial\Psi}\Delta\Psi^\dagger + \frac{\partial F_j^\dagger}{\partial\Psi}\Delta V_j + \frac{\partial G^\dagger}{\partial\Psi}\Delta\Psi + \Delta\dot{\Psi}^\dagger, \delta\Psi\right)_{\Gamma\times\mathcal{T}} \\
&-\left(\left(\Psi^\dagger\frac{\partial^2 G}{\partial p_1\partial V_j} + V_i^\dagger\frac{\partial^2 F_i}{\partial p_1\partial V_j}\right)\Delta p_1 + \frac{\partial F_i}{\partial V_j}\Delta V_i^\dagger + \frac{\partial G}{\partial V_j}\Delta\Psi^\dagger + \frac{\partial F_i^\dagger}{\partial V_j}\Delta V_i + \frac{\partial G^\dagger}{\partial V_j}\Delta\Psi - \Delta F_j^\dagger, \delta V_j\right)_{\Gamma\times\mathcal{T}}
\end{aligned} \tag{20}$$

Note that the variations of $F_j^\dagger$ and $G^\dagger$ can be unpacked further, if desired:

$$\delta F_j^\dagger = \underbrace{\left(\frac{\partial^2 F_k}{\partial V_j\partial V_i}V_k^\dagger + \frac{\partial^2 G}{\partial V_j\partial V_i}\Psi^\dagger\right)}_{\frac{\partial F_j^\dagger}{\partial V_i}}\delta V_i + \underbrace{\left(\frac{\partial^2 F_k}{\partial V_j\partial\Psi}V_k^\dagger + \frac{\partial^2 G}{\partial V_j\partial\Psi}\Psi^\dagger\right)}_{\frac{\partial F_j^\dagger}{\partial\Psi}}\delta\Psi, \tag{21}$$

$$\delta G^\dagger = \underbrace{\left(\frac{\partial^2 F_j}{\partial\Psi\partial V_i}V_j^\dagger + \frac{\partial G^2}{\partial\Psi\partial V_i}\Psi^\dagger\right)}_{\frac{\partial G^\dagger}{\partial V_i}}\delta V_i + \underbrace{\left(\frac{\partial^2 F_j}{\partial^2\Psi}V_j^\dagger + \frac{\partial^2 G}{\partial^2\Psi}\Psi^\dagger\right)}_{\frac{\partial G^\dagger}{\partial\Psi}}\delta\Psi. \tag{22}$$

The variation vanishes if the second-order adjoint solution satisfies the problem below.

Second-order adjoint problem

$$\begin{aligned}
\rho\Delta\ddot{u}_J^\dagger &= \partial_I C_{IJKL}\partial_K \Delta u_L^\dagger + \Delta Q_J^\dagger, && \overline{x}\in\Omega, && t\in\mathcal{T},\\
\Delta\overline{u}^\dagger &= \overline{0}, \quad \Delta\dot{\overline{u}}^\dagger = \overline{0}, && \overline{x}\in\Omega, && t=T,\\
L^\dagger\Delta\overline{u}^\dagger &= 0, && \overline{x}\in\partial\Omega, && t\in\mathcal{T},\\
\Delta V_n^\dagger &= 0, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\overline{T}^{\dagger+} &= -\Delta\overline{T}^{\dagger-}, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\tau_J^\dagger &= \Delta F_J^\dagger, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
-\Delta\dot{\Psi}^\dagger &= \Delta G^\dagger, && \overline{x}\in\Gamma, && t\in\mathcal{T},\\
\Delta\Psi^\dagger &= 0, && \overline{x}\in\Gamma, && t=T.
\end{aligned} \tag{23}$$

where

$$\begin{aligned}
\Delta F_j^\dagger &= \frac{\partial G^\dagger}{\partial V_j}\Delta\Psi + \frac{\partial F_i^\dagger}{\partial V_j}\Delta V_i + \frac{\partial G}{\partial V_j}\Delta\Psi^\dagger + \frac{\partial F_i}{\partial V_j}\Delta V_i^\dagger\\
&\quad + \Delta p_1\left(\Psi^\dagger\frac{\partial^2 G}{\partial V_j\partial p_1} + V_i^\dagger\frac{\partial^2 F_i}{\partial V_j\partial p_1}\right),\\
\Delta G^\dagger &= \frac{\partial G^\dagger}{\partial \Psi}\Delta\Psi + \frac{\partial F_j^\dagger}{\partial \Psi}\Delta V_j + \frac{\partial G}{\partial \Psi}\Delta\Psi^\dagger + \frac{\partial F_j}{\partial \Psi}\Delta V_j^\dagger\\
&\quad + \Delta p_1\left(\Psi^\dagger\frac{\partial^2 G}{\partial \Psi\partial p_1} + V_j^\dagger\frac{\partial^2 F_j}{\partial \Psi\partial p_1}\right)
\end{aligned} \tag{24}$$

$$\Delta Q_j^\dagger = \sum_{k=1}^{N_{rec}} \Delta\dot{u}_j\hat{\delta}(\overline{x}-\overline{x}_r^{(k)}) \tag{25}$$

Just like the adjoint problem, the second-order adjoint problem is solved in reversed time. It is identical to the adjoint problem up to source terms. It has both a wavefield source $\Delta Q_j^\dagger$ and source terms in the friction law and state evolution. The wavefield source $\Delta Q_j^\dagger$ contains the second order forward wavefield $\Delta\dot{u}_j$ injected at the receivers, similar to how the forward wavefield is injected in the adjoint problem. The coefficients in the friction law and state evolution equation depend only on the forward solution (just like for the adjoint problem), but the source terms also depend on the adjoint solution and the model perturbation. Thus the second-order adjoint problem depends on all three previous problems and must be solved last.

Summary: To compute the Hessian applied to a perturbation Δp_1, we need to solve the FP, AP, second-order FP and second-order AP, and then evaluate (10).

Note that we assumed no filtering for simplicity, i.e. $W[x] = x$, which makes the variation of $Q_j^\dagger$ easy to compute.

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